

TWST · FY2026 Q2

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Twist Bioscience Corporation · 47 turns · 16 participants

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Welcome to Twist Biosciences' 2026 Second Quarter Financial Results Conference Call. At this time, all participants are in listen-only mode. After the speaker's presentation, there will be a question and answer session. To ask a question, you will need to press star 1-1. In the interest of time, we ask that you please limit yourself to one question. Also note, this call is being recorded. I would like to turn the call over to Angela Bidding, SVP of Corporate Affairs. Please go ahead.

ANGELA BIDDING SVP of Corporate Affairs

Thank you, Operator. Good morning, everyone. I would like to thank you for joining us for TWIST Bioscience's conference call to review our fiscal 2026 second quarter financial results and business progress. We issued our financial results press release before the market, and it is available at our website at www.twistbioscience.com. With me on the call today are Dr. Emily LaPruce, CEO and co-founder of TWIST, Adam Loponis, CFO of TWIST, and Dr. Patrick Finn, President and CEO of TWIST. Today we will discuss our business progress, financial and operational performance, as well as growth opportunities. We will then open a call for questions. We ask that you limit your questions to only one and then re-queue as a courtesy to others on the call. This call is being recorded. The audio portion will be archived in the investor section of our website and will be available for two weeks. During today's presentation, we will make

forward-looking statements within the meaning of the U.S. federal securities laws. Forward-looking statements generally relate to future events or future financial or operating performance. Our expectations and beliefs regarding these matters may not materialize, and actual results and financial periods are subject to risks and uncertainties that could cause actual results to differ materially from those projected. These risks include those set forth in the press release we issued earlier today, as well as those more fully described in our filings with the Securities and Exchange Commission. The forward-looking statements in this presentation are based on information available to us as of the date hereof, and we disclaim any obligation to update any forward-looking statements, except as required by law. We'll also discuss adjusted EBITDA, a financial measure that does not conform with generally accepted accounting principles. Information may be calculated differently than similar non-GAAP data presented by other companies. When reported, a reconciliation between GAAP and non-GAAP financial measures will be included in our earnings documents, which can be found on the investor section of our website. With that, I will now turn the call over to our CEO and co-founder, Emily Leproust.

DR. EMILY LEPROUST CEO & Co-founder

Thank you, Angela, and good morning, everyone. To which delivered another strong quarter, I extended our track record of consistent execution, posting our 13th quarter of sequential revenue growth. We have outperformed the broader lifetime tools market with a model that scales efficiently and drives increasing value creation. Twist Core Technology Advantage is a semiconductor-based DNA synthesis platform that provides a structural advantage in cost, scale, and speed that feeds into every product and service we offer. The same platform also enables a highly efficient new product introduction engine allowing us to rapidly translate customer demand into scalable offerings and continuously expand our portfolio. As we increase volume on the silicon chip, we expand our wallet share, accelerate product innovation, and further strengthen our competitive advantage. The model works exactly as designed. We have delivered sustained revenue growth, expanded gross margin above 50%, invested strategically to drive continued return on that investment, and we remain firmly on track to achieve HSBC beta break-even in the first quarter of fiscal 2026. Focusing our results for the second quarter of fiscal 2026, we grew total revenue to \$110.7 million, up more than 19% year-over-year. DNA synthesis and protein solutions grew 28%, powered by continued strength in AI-enabled drug discovery. NGS applications grew 12% year-over-year and 9% sequentially. Studying deeper into DNA synthesis and protein solutions, we continue to see robust growth. Last month, Amazon Web Services announced Twist as a web lab partner for Amazon BioDiscovery, its AI-powered drug discovery application. This is an exciting validation of our DNA synthesis, protein solutions, and biologics capabilities. In advance of the launch, Twist has been working with AWS team for several months, providing web lab services for the applications scientific launch partners, including Memorial Sloan Kettering Cancer Center and the Gray Lab at Johns Hopkins University. The objective for researchers using Amazon BioDiscovery is to deploy AI models to design and optimize antibody candidates faster. We're here to support them with products and services that accelerate that pathway. By staying in close contact with our customers, we identified this emerging category of AI-enabled drug discovery early, and invested ahead of the market acceleration, with increasing adoption across pharma, dry lab, and big tech companies. Importantly and unbalanced, the growth of AI-enabled rediscoveries complements our work with customers pursuing traditional rediscovery, which remains a robust area of our business. Regardless of approach, Our customers for DNA synthesis and protein solutions are all working through the same fundamental design-build-test-learn cycle. What differs is how they execute against that framework. There is no one-size-fits-all model. Each program is tailored to the customer scientific approach, resources, and stage of discovery. What remains constant across every engagement is the foundation, twist silicon platform, which enables cost-effective synthesis of hundreds of thousands of unique sequences in parallel. That unique and proprietary capability is what makes speed at scale possible, no matter where the customer enters the workflow. No two orders are identical, so we see consistent patterns in how

these campaigns are structured. On slide six, to give you some context, one example is our work with Memorial Sloan Kettering and Amazon. where the team ordered approximately 100,000 specific DNA sequences as a pooled library. This approach is highly efficient precisely because of how our platform is built. Pooled DNA can be manufactured collectively rather than individually cloned and processed, driving down curves per sequence dramatically. And with this DNA provider, we can deliver hundreds of thousands of specific sequences pooled at speed and scale. Once they have the pooled DNA, either twist or the customer then screens that library to identify promising candidates, selects the most relevant sequences, and advances those into individual synthesis, protein expression, and characterization. Through iterative cycles, this process yields validated antibody leads. The second model involves customers ordering hundreds to thousands of gene fragments and executing downstream workflow internally. Here again, TWIST platform delivers an edge in the ability to synthesize diverse sequence sets quickly and at accessible cost, meaning customers can explore broader design spaces. In these cases, customers convert fragments into clonal genes, express proteins, and perform characterization assays within their own laboratories. Others choose to start further downstream, purchasing clonal genes or antibodies and binding proteins, such as IgG, SCID, DHH, and others, to focus their internal efforts on functional characterization and validation. Even at this entry point, the advantage traces back upstream. The parallel synthesis capability underpinning our platform ensures the sequences they receive reflect the speed at scale that alternatives cannot match. And we have a growing segment of customers we rely on as an end-to-end partner. In these engagements, We handle DNA synthesis, clonal gene constructions, protein expression, and characterizations. Our platform's ability to run large, complex sequence sets in parallel accelerates every stage of that workflow, and we deliver high-quality experimental data that enables customers to focus on critical analysis, decision making, and iterative design. We also have a number of customers who give us a biological target and ask us to do all of the work through individual, in vitro, and our AI ML discovery approaches. Across all of these models, Curve scales with the scope and complexity of the workflow, ranging from smaller exploratory programs to multi-million dollar discovery efforts. Our role is to provide flexibility across the spectrum. Because our platform was purpose-built for parallel synthesis at scale, we can meet customers where they are, whether they need a pool of libraries of hundreds of thousands of sequences or a fully managed discovery program. We support them as their research advances. On slide 7, you'll see our portfolio for DNA synthesis and protein solutions serving customers across the biological continuum. Building on our success in serving therapeutic discovery customers, in February, We license the B-body bispecific platform to expand our capabilities in this rapidly growing modality. We now enable high throughput discovery in bispecifics, an area that has strictly been limited by scale. We have already received our first orders for this platform with a robust funnel looking forward. Moving to slide 8, NNGS grows re-accelerated in the second quarter. Our NGS tools business remains a durable and growing part of the portfolio with particular strengths in oncology diagnostics. We operate at the critical part in the workflow between the sample and the sequencer, where our products support precision and customization at scale. Our target enrichment and laboratory preparation solutions deliver the uniformity and on-target performance required for high-sensitivity applications. This is especially relevant in the continuum of cancer care on slide 9, where we are seeing increasing adoption in commercial diagnostic tests, including initial momentum in molecular or minimal residual disease or MRD. These applications demand extremely high accuracy and reproducibility, fast and on time, and our chemistry is well aligned to these requirements. Specifically, on slide 10, as MRD testing transitions from early clinical adoption to scale deployment across oncology diagnostics, the technical and operational requirements become significantly more demanding. These assays are pushing the limit of sensitivity, often requiring detection of variants at extremely low allele frequencies. That places a premium on panel design as well as the entire workflow to ensure uniform coverage and reproducibility across REM. In this environment, success depends on the ability to deliver highly customized target enrichment panels, laboratory preparation enabled by novel enzymes, as well as the buffer, BCDI, UMIs, and other components optimized for specific indications and evolving clinical needs. Equally

important is speed. As these tests move into broader clinical workflows, laboratories, and diagnostic developers need a rapid turnaround on panel design, synthesis, and deployment to support asset development timelines and commercial scale-up. At Waze, we combine high-throughput DNA synthesis with precision probe design and manufacturing at scale, enabling fast, reliable delivery of customized panels with consistent performance characteristics. That allows our customers to move quickly from development to validation and commercialization without compromising on data quality, and importantly, securing and future-proofing their stable supply chain. For bespoke or tumor-informed MRD panels, like all of our NGS panels, this is a consumer-driven workflow that scales with state volume, supporting recurring revenue as these applications expand. This time, I'd like to turn the call over to Paddy to expand further on our growth initiative around the coming product offering.

DR. PATRICK FINN President & CEO

Thank you, Emily. Halfway through our fiscal year, the results are strong and we believe the road ahead is stronger. Everything we do in protein solutions and AI-enabled discovery runs on one foundation, our DNA synthesis platform. It's a structural advantage for cost, scale, and speed, full stop. And we continue to advance and strengthen this platform to enhance customer experience even more. Today, we accept the vast majority of daily sequences as we know we can manufacture them. We have an algorithm embedded in our e-commerce system to inform a customer immediately if they have uploaded a sequence that may be difficult to manufacture. This notification improves the user experience for customers as some sequences present manufacturing challenges. Repeat regions, hairpins, extreme GC content. Three years ago, we accepted about 96% of clonal genes and could manufacture about 97.5% of clonal genes and about 98% of DNA sequences more broadly, including oligopools, DNA libraries, gene fragments. Today, we accept about 97% of clonal genes and can manufacture approximately 98.5% of clonal genes and about 99% of DNA requests more broadly. That's not theoretical capability. That's production reality at scale. We routinely deliver clonal genes and fragments up to 5,000 base pairs, oligopools up to 300 bases, multiplex gene fragments up to 500 base pairs across a wide range of formats. If a customer can design it, we are increasingly able to make it. Even as an acceptance rate for DNA sequences remains very high, we recognize that when a single sequence in a larger set does not meet acceptance criteria, customers may choose to route the full set elsewhere. This dynamic highlights a clear opportunity. Continued improvements in acceptance rates can unlock incremental share gains and expand total order capture. On slide 11, You'll see that we announced this morning that we will soon take a full range of sequences across length and complexity, driving toward accepting approximately 99.5% of clonal genes and 99.9% of all DNA products more broadly. This constant drive to improve sets us apart, and importantly, more sequences accepted means more orders won, and we intend to win them. And that matters, because this is not a standardized market. Every customer is different, Every order is different. Bread drives share game. It's that simple. In contrast to Twist, the competitive landscape has a pattern. Niche players, narrow offerings, limited reach. That is not how customers operate, and it is not how this market is won. We employ a different approach. We anchored our strategy around end applications, where performance, scale, and execution are the only metrics that matter. Customers do not want to stitch together multiple vendors. They want one partner who can deliver consistently across a workflow. That is where Twist is differentiated. We offer both depth and breadth across the biological continuum, from DNA synthesis through proteins, biologics, and NGS. We support customers increasingly across the entire lifecycle of their programs. This is how we continue to take share, expand wallet, and reinforce our positions in leading platforms serving therapeutics, diagnostics, industrial, academic, and government markets. With that, I'll turn it over to Adam to discuss our financials for the quarter.

ADAM LOPONIS Chief Financial Officer

Thank you, Patty. Turning to slide 12, Q2 is another quarter of consistent execution against the financial model we've laid out. Revenue grew 19.3% year-over-year to \$110.7 million our 13th consecutive quarter of sequential growth. Gross margin expanded to 51.6 percent versus the prior year, an improvement of approximately 200 basis points. And we remain firmly on track for adjusted EBITDA breakeven in Q4. Let me walk you through the details. On slide 13, you'll see DNA synthesis and protein solutions revenue increased to 53.3 million, growth of 28 percent year over year. On slide 14, We show NGS applications revenue for the second quarter grew to approximately \$57.4 million compared to \$51.1 million in the second quarter of fiscal 2025. An increase of 12% year-over-year and up 9% sequentially, driven by growth in top accounts. For the quarter, revenue from our top 10 NGS applications customers accounted for approximately 39% of NGS applications revenue. We serve 627 NGS applications customers in the quarter, with 174 having adopted our products. Looking geographically on slide 15, America's revenue increased to approximately \$64.3 million in the second quarter, compared to \$55.2 million in the same period of fiscal 2025, growth of 17% year over year. EMEA revenue rose to \$37.3 million in the second quarter, versus \$30.6 million in the same period of fiscal 2025, growth of 22% year-over-year. APAC revenue increased to \$9.1 million in the second quarter compared to \$7 million in the same period of fiscal 2025, an increase of 30% year-over-year. APAC accounted for 8% of our revenue in the second quarter. China continues to be a relatively small portion of our revenue at approximately 1% of total revenue for the second quarter of fiscal 2026. Looking at revenue by industry on slide 16. Therapeutics revenue rose to \$40.8 million for the second quarter of 2026, compared to \$26.3 million in the same period of fiscal 25, growth of 55%, reflecting the increased uptake of our products by large pharma and biotech customers in their efforts on therapeutics discovery and including AI-enabled discovery. Diagnostics revenue was \$40 million for the second quarter of 2026, compared to \$35 million in the same period of fiscal 2025, an increase of 14%. Diagnostics revenue grew 13% versus Q1 of fiscal 26, based on strong growth from top accounts. Industry and applied revenue was \$5.8 million in the second quarter of 26, compared to \$7 million in the same period of fiscal 2025. Academic research and government revenue was \$12.8 million for the second quarter of 26, compared to \$12.5 million in the same period of fiscal 2025, It increased to 3%. Sequential growth was 5% versus prior quarter, driven by strength in U.S. accounts. Global supply partner revenue was \$11.4 million in the second quarter of 2026, compared to \$12 million in the same period of fiscal 2025, primarily due to order timing. Moving down to P&L on slide 17. Our gross margin for the second quarter increased to 51.6%, an improvement of two margin points versus the same period of fiscal 25. Margin expansion was driven by strong revenue growth and moderated sequentially as we continue to invest in new product offerings and manufacturing capacity that we expect will result in future margin gains as we accelerate growth and implement continuous process improvement. Operating expenses, excluding cost of revenues and litigation settlement costs, or \$95.8 million for the quarter, compared to \$87.6 million in the prior year. The increase reflects deliberate investment in our commercial organization and digital infrastructure to support the growth trajectory we are delivering, particularly the 55% growth in therapeutics. These are revenue-generating investments with a clear line of sight to return. We are managing these investments with discipline. In April, we reduced 36 positions to reallocate resources to our highest return opportunity. Combined with additional cost initiatives underway, we expect these actions to contribute to sequential OPEX improvement of \$6 million in Q4 of fiscal 26. Looking at our progress on our path to profitability. In the second quarter of fiscal 2026, adjusted EBITDA was a loss of approximately \$13.3 million, an improvement of approximately \$1.5 million versus the second quarter of fiscal 25. we have systematically narrowed that loss through a combination of revenue growth, gross margin expansion, and operating expense discipline. We expect the actions we've taken, combined with continued revenue momentum, to fully deliver on our targets for Q4. We reached an agreement in principle regarding the securities class action for approximately \$17.1 million. In fiscal Q2, we booked \$7.2 million for litigation settlement costs net of recoveries as we expect the additional cost to be covered by our insurance. We view this as a positive resolution allowing management to remain fully focused on execution. We ended Q2 with \$171.7 million in cash, cash equivalent

ents, and short-term investments versus \$197.9 million as of December 31, 2025. Sequential change reflects \$17.6 million in operating cash usage \$7.9 million in CapEx as we continue to invest in manufacturing automation, and \$5 million in cash for the Inventora license and equity investment. On slide 18, turn to guidance. For fiscal 2026, we expect total revenue of \$442 million to \$447 million, growth of approximately 17% to 19%. For Q3 of fiscal 2026, we expect total revenue of \$114 million to \$115 million, growth of approximately 19% year-over-year at the midpoint. As previously discussed, we expect NGS to be the driver of sequential growth in H2 and return to 20% growth by Q4. We remain confident in our trajectory and continue to forecast reaching adjusted EBITDA breakeven for the fourth quarter of fiscal 2026. With that, I'll turn the call back over to Emily.

DR. EMILY LEPROUST CEO & Co-founder

Thank you, Adam. On slide 19, as we look ahead we remain focused on delivering consistent measurable growth designed to scale over time. We see strong momentum across the portfolio, with continued growth in DNA synthesis and protein solutions, increasing adoption in AI-enabled discovery, and a return to growth in MGS. We serve large and expanding markets where our platform is increasingly relevant. At the same time, our operating model continues to perform as expected. We have delivered 13 consecutive quarters of sequential revenue growth expanding growth margins above 50% and maintaining a clear path to a GCDB data break-even in fiscal 2026. What underpins this performance is the scalability of our platform. As volume increases, we expand our ledger, improve efficiency, and generate operating leverage across the business. Our ability to serve a wide range of customer workflows from early discovery through clinical and diagnostic applications provides both resilience and opportunity to capture more value over time. Across the business, we invest with discipline. We target high return opportunities, allocate capital deliberately, and align investments with clear growth drivers. We execute with focus and urgency to drive durable growth and build a company with increasing strategic relevance and long-term value creation. With that, we're happy to take your questions. Operator?

OPERATOR

Thank you. As a reminder, to ask a question, please press star 1-1. If your question has been answered and you'd like to move yourself in the queue, please press star 11 again. We also ask that you please limit yourselves to one question. Our first question comes from Mack Etoc with Stevens. Your line is open.

MACK ETOC Stevens & Co Analyst

Hey, good morning, and thank you for taking my questions. Maybe just to start, could you just discuss how AI-driven workflows performed in the quarter relative to your internal expectations and how the change in the outlook for both of the segments is really contributing to the change in the updated fiscal guidance from here. Thanks.

DR. EMILY LEPROUST CEO & Co-founder

Yeah, thanks for the question. Obviously, we're very excited with the performance of the SPS, growing 28% year-over-year. And for the therapeutic category, we cracked the \$40 million for the quarter. A lot of companies in the drug discovery field, they tap out at \$50 million a year. And now we're way past that. We're almost there every quarter. So obviously, there's strength. There's strength throughout the menu. What we see is that customers don't want one thing. And so the NTI engine that we built that creates a lot of options for people to enter has been a great driver. And obviously, AI-driven discovery has been a big help in that area. It just increases the number of sequences that people want to look at. If they looked at 100 sequences before, now with AI, they can get thousands. And so it just increases the overall value of the deals. And a lot of companies don't have the capacity to analyze that number of antibodies. And so it enables us to upscale, upsell to data and cell characterization. Overall, the entire menu is doing well, but air hydrogel discovery has definitely been a great win in ourselves.

OPERATOR

Thank you. Our next question comes from Vijay Kumar with Evercore ISI. Your line is open.

VIJAY KUMAR Evercore ISI Analyst

Hi, Emily and Adam. Good morning to you, and thank you for taking my questions. Congrats on a nice sprint. Maybe on the prior question related to it, when I look at therapeutics, up 55% in second quarter, that's an acceleration from Q1 growth levels. How much of this acceleration was driven by AI-related programs? I think in the past, Emily, you've called it data characterization genes versus traditional biopharma programs. And when you look at back half, is this 50% kind of growth sustainable when you look at your order book and backlog? Thank you.

DR. EMILY LEPROUST CEO & Co-founder

That's a great question. Definitely, AI has been a source of strength. Again, the nice thing as people develop more sequences, whichever levels of DNA we can make, if they want a pool library of VHH, we can make it. If they want, even for by specific now, we can make gene pools. We can make the flavor of the DNA unpooled. So the entry point of whatever they want, that broad menu is very useful. But with AI, as I mentioned earlier, there's more need for data characterization. And what has happened is maybe in 2025, there was excitement, definitely on our side, but that excitement came from a very few number of accounts. And now that we are many quarters into this, now it's dozens of accounts that are driving that growth, right? It's not just a few. And so we can see that not only it's repeatable with the existing account, but we're able to bring more and more people into the fold. And then sometimes we can enter through AI-driven discovery, but a lot of those companies are doing both AI-driven discovery and traditional discovery. And the fact that we have a full menu enables us to grow in all areas. So overall, it's growth-based. Going forward, we're not guiding per product groups. We think there's very good growth potential within the business. And we doubled growth. The raise is double the beat, right? So obviously there's a lot of confidence, but that confidence is broad-based. And we also share some strong confidence for our NGS business. So the business is ripping. The sales team is confident. Customers are happy.

UNKNOWN

So we just have to do it again.

OPERATOR

Thank you. Our next question comes from Doug Schenkel with Wolf Research. Your line is open.

DOUG SCHENKEL Wolf Research Analyst

Good morning. Thank you for taking my questions. I want to just cover two topics, the academic and government end market, and then gross margin. So in A&G, what are you seeing? Are things stabilizing or improving? Just want to get a sense for how things are trending. And are you still running the academic promotion on express jeans? Kind of building off of that, when do we start to lap the headwinds on price per gene from expressed gene fragments? I think that's this summer, so just want to be mindful of that as we're updating our models. And then a quick one on gross margin. Gross margin was down nominally, sequentially, in a little bit light of our model and street models. That just may be in the noise, but wanted to see if there's anything to call out there. Thank you.

DR. EMILY LEPROUST CEO & Co-founder

Yeah, so maybe I'll take the first question and Adam will cover the second one. Yeah, on academic, definitely that end market is suffering from funding pressures. Our approach is to take market share. And so academics, when funding is under pressure, they you know, they're very cautious about their dollars. And so, especially, you know, that

market is basically shrinking right now. And so the fact that we're growing and growing sequentially is definitely a good thing. It shows that our product offering resonates more than the competition and so right now we are very happy to extend the premium discount for what that enables academic people to get X-rayed genes at the price of standard genes and so they get a great value on on the DNA they're receiving from us. They're getting great speed. It enables them to go faster and get better at their next grant, and knowing that those grants are very competitive. So I think our discount is very well received by those customers. And definitely, the growth there is smaller than for an industry segment, but we are taking market share. Adam, you want to cover the gross margin question?

ADAM LOPONIS Chief Financial Officer

Absolutely. Welcome, everyone. And Q2 definitely reflects a deliberate investment, specifically for IGG and characterization for AI discovery projects, as well as our digital capabilities. We remain confident in our 52% for better guide for the year. And this investment is really around adding capacity of people to support the accelerated demand. We see a huge ROI on it, and we also see the path to continuously making improvements on those efforts on new products and returning to a 75% to 80% average drop through on incremental revenue to gross margin as we move forward and automate workflows.

OPERATOR

Thank you. Our next question comes from Luke Sergat with Barclays. Your line is open.

LUKE SERGAT Barclays Analyst

Thanks. Just a couple here. I want to follow up on Doug's question there. Adam, when you're talking about the gross margin improvement and the automated workflows, you guys have just built out the new facility. It's pretty state-of-the-art from what you had previously. Talk about further investment or how much more you guys can continue to push that automated workflow. Then I wanted to follow up and ask more on the data characterization of the AI projects. You guys talked about \$25 million in bookings in 4Q coming from some of these AI projects. How much of the revenue in the first six months of the year has been converted from that? Or is that still majority on the come?

DR. EMILY LEPROUST CEO & Co-founder

Thanks for the question. So maybe I'll start with the first one. And you guys are really good at squeezing two questions into one at the same time. Yeah, I see what you're doing there. So on the first one, if you remember conceptually, our front end and the back end. So the front end is the silicon chip. That's where we make the oligos. That's where we get the massive advantage. And then after the logos are made, depending on the flavor, it goes to different backends. So in general, the backend for our DSPS is a different backend than for NGS. And then in DNA synthesis in a protein solution, we first make a fragment, and then some stop there, they get shipped. Then we make clonogenes, some stop there, they get shipped. And then we make IgG expressions. And then some stop there, they get cheap, and then some of the agility gets characterized. And so as you can imagine, as we add flavor, we have to add a little bit of automation to add capacity for that new flavor. So it's not a tremendous amount of capex. It's significant, but compared to building a new fab from scratch, it's not comparable. And so that's the strategy we've been using is following the demand from customer and adding a little bit of automation on the backend, on the branch that's needed. So that's one. And then the other thing we've done is we've been automating the automation. And so for those of you that come to our investor day, you'll see that there was a giant room Well, now in that one room, we've been able to automate the automation, and now we'll be able to have multiple times the capacity that we used to have in that room. In terms of your second question around the \$25 million of orders that we had last year, those have all been shipped now. Most of that was shipped online. uh was finished uh by q1 and so there's very little impact of of that um in in q2 as you remember um one thing that's that's a key differentiation from us is the speed at

which we deliver data and so you know some data most of the data is delivered 15 to 20 days after we receive the sequence so it's a quick turnaround to to book the rubbish

OPERATOR

Thank you. Our next question comes from Sabu Nambi with Guggenheim. Your line is open.

SABU NAMBI Guggenheim Analyst

Hey, guys. Thank you for taking my questions. You increased your full-year revenue guidance by more than the magnitude of the Q2 beat. Any specific area or areas which drove the increase? Essentially, I'm asking for the bridge. Thank you.

ADAM LOPONIS Chief Financial Officer

Hi, Thibu. Great to hear you. Yes, no, what we talked about in the back half, while sequentially most of the improvements are going to come from NGS, if you look at the full year guide, of the raise in the back half. So if you are modeling it out sequentially, we don't expect anything to go backwards in terms of DSPS. We only expect continued increases in the overall revenue, but at a more modest rate. And again, it really is dependent upon the pace of us getting new customers into the DSPS side and the ability to potentially exceed those expectations is an area of opportunity for us.

OPERATOR

Thank you. Our next question comes from Matt LaRue with William Blair. Your line is open.

MATT LARUE William Blair Analyst

Hi, good morning. I wanted to ask on the complex DNA offering, you know, what's the timeline at which you expect to deliver products sort of in line with the the capability you described today, what lengths do you expect to be able to get up to in terms of manufacturing? and do you have any sense or can you give us a sense or you know what the missed opportunity has been historically um you know you referenced in some in some cases losing a whole order because you couldn't make a sequence so what what kind of additional opportunity does this unlock by adding these new capabilities thanks hey hey matt it's petty here thanks for the question um yeah we're pretty excited about the product i think you could hear from uh

DR. PATRICK FINN President & CEO

My comments are actually we accept, you know, the vast majority of sequences that come our way across. All of the portfolio, and it is a few percentage or a few percentage points of sequences. We don't take. Which we see some instances where the customer wants to take that total order somewhere else. The trends that are emerging nucleic acid, therapeutics and plant engineering and of course, AI. You know, they have some special needs or special requirements around the sequence complexity. And so, you know, in true twist fashion, we're focused on the customer experience. Our goal is to create a one-stop shop. You know, and when you leverage your DNA synthesis platform that can print really high-quality DNA, and just as a reminder, you know, up to approximately \$6 billion per day, that gives us super quality speed and economics at scale, including for complex sequence. And so now with a little bit more optimization throughout the entire workflow, it's going to deliver a product that's best in class. So broad sequence acceptance, really strong, predictable, and transparent performance when you're on the Twist platform. Twist reliability, our platform's industrialized. It's faster, cheaper, strong best-in-class e-com user experience. Then you've got Twist customer service and support, So that's trained scientists that understand the customer's experiment. They are there to help the customer through any of their challenges. So there's a lot to like with the offering.

And we're looking forward to scaling in the coming few quarters. So we're talking about early access. Let's start with this classic twist, make a few customers happy and learn as we go. And you'll continue to see the capability ramp up as we go through the quarters.

OPERATOR

Thank you. Our next question comes from Catherine Schultz with Baird. Your line is open.

CATHERINE SCHULTZ Baird Analyst

Hey, everyone. Thanks for the question. Maybe on the margin side, still on target to hit adjusted EBITDA breakeven in the fourth quarter, you've been very prescriptive about gross margin incrementals. I guess as you hit that milestone, how should we think about leverage beyond that point and maybe EBITDA margin incrementals going forward?

ADAM LOPONIS Chief Financial Officer

Hi, Gavin. Good morning. Thank you for the question. In terms of where we are today, we're laser focused on making sure we cross the adjusted EBITDA positives here at Q4, while also at the same time ensuring to do it at the maximum acceleration of revenue growth. So it's really titrating that investment in such a way that we maximize revenue growth rates. As we go forward, we have optionality. We talked about that before. We'll talk about it more in the future. But we're stewards of the market and we understand that our continued path of not going backwards is very important to us and also ensuring that we continue to sustain the accelerated levels of growth. So those will be the two focus areas. And I think we'll spend some more time talking about that at our investor day here coming up in May.

OPERATOR

Thank you. Our next question comes from Puneet Sudha with Lyric Partners. Your line is open.

PUNEET SUDHA Lyric Partners Analyst

Hi, everyone. You have Michael on for Puneet. Congrats on the quarter. I was hoping to get some color on the genes, so I saw a strong growth in the physical gene shift. Last quarter, you talked about 50K for characterization. I was wondering if you could offer any color on the contribution of genes for characterization this quarter. And if you could offer any insight to us on how much of this AI demand is driven by more model building versus incorporation of AI into your ongoing drug discovery lead generation work at pharma.

DR. EMILY LEPROUST CEO & Co-founder

Thank you. Great question. We did not share the number today. We're trying to straddle the fine line between being as transparent or as possible and not tipping of the competition too much. I'm sure some of them are on the line today. What we can say is that it's more. So there's definitely growth in how many genes were used internally to generate data that was sold. So definitely

UNKNOWN

So definitely there is growth there.

DR. EMILY LEPROUST CEO & Co-founder

And the second, there was a second question.

ADAM LOPONIS Chief Financial Officer

Yeah, it was around the ability to, how much of the growth is coming from making the models?

DR. EMILY LEPROUST CEO & Co-founder

Oh, yeah. Thanks. I'm sorry, I was focusing on the number. Yeah, very interesting, actually. We've seen a lot of customers that have shifted. While the beginning was a bidding model, I think by now most of them, or a lot of them, are now turning the crank. And so when they turn the crank, what we see is that The others, maybe each order, maybe a little bit smaller. So building a model, it's a big bonus upfront. So it's a little bit smaller, but it's more often. And so we're very pleased to see that people are returning. And yeah, it's working as expected.

UNKNOWN

It's working well.

OPERATOR

Thank you. Our next question comes from Brendan Smith with TD Cowan. Your line is open.

BRENDAN SMITH TD Cowen Analyst

Great. Thanks for taking the questions, guys. Maybe just to put a pin in the AI orders questioning here fully. I guess looking at that \$25 million from last year, do you have a sense or can you tell us what kind of run rate you're looking at now for FY26, maybe just with the first half under your belt? And I guess related to that and within those revenues, I guess, can you give us a sense what the relative breakdown is between revs from, you know, oligos versus IgG versus analytical data, just kind of wondering really how much of the AI revs are mostly done kind of farther down that funnel or really where they're shaking out. Thanks.

ADAM LOPONIS Chief Financial Officer

Brandon, I can start with a little bit of color around how to think about where we're going. If you think about the therapeutic segment as a whole, obviously the AI discovery is falling in there. And if you look at the outsized growth versus the average, the business, that delta is predominantly AI discovery work. I will say at times it's hard to know exactly whether it's, you know, discovery or classic work, we don't always know exactly the number, as well as, you know, whether it's training or building the models. But pretty strong confidence that the vast majority of the outside growth in therapeutics is AI discovery.

UNKNOWN

I'm going to hand it to you in a second. Yeah, go ahead.

OPERATOR

Thank you. Thank you. Our next question comes from Tom DeBorsi with Nefron Research. Your line is open.

TOM DEBORSI Nefron Research Analyst

Thanks for taking the question. Diagnostics as a whole and the, you know, double-digit growth and also double-digit sequential growth is sort of you had, I guess, projected before. Just as you think about the rest of the year, do you see incremental sequential growth through Q3, Q4, just thinking about diagnostics customers and their contribution to NGS?

DR. EMILY LEPROUST CEO & Co-founder

Thank you. Thank you, Tom. Yeah, definitely. We're seeing growth. I think in Q4, we're actually seeing at least 20% growth in NGS alone. And it's a good reminder. You know, we talked a lot on the call about DNA synthesis and protein solution and a hydro discovery. You know, we love it all. At the same time, it's a good reminder that the dollar growth in NGS or dollar growth in instances and protein solution is very similar to us. And so what we're optimizing

for as a management team It's for revenue growth for the entire business, growth margin above 50% and getting to a JCV tab break even. And the portfolio of panels that are still library prep that we build, in the NGS application group enables us to sustain great growth. We are liquid biopsy customers, are MRD customers, are ramping and adopting. And so we are definitely very confident in that part of the business. And we're very much looking forward to return to growth. Not so long ago, it was flipped where we had, you know, eight-ish percent growth in NGS and 12% growth in DNA synthesis and protein solution, it's going to flip-flop back and forth. But at the end of the day, what matters to us is the entire business growth. If you look back into the you know life science tools uh industry um hopefully we we are unique in the the kind of goals that that we are uh um being able to to post um not just this quarter but you know by now certain quarter uh in in a row um so uh 110.7 this quarter and not so long ago um we had \$90 million for the entire year, right? So we'll keep doing it, and I think the future is very bright.

OPERATOR

Thank you. There are no further questions at this time. I'd like to turn the call back over to Emily LaPruce for closing remarks.

DR. EMILY LEPROUST CEO & Co-founder

As we wrap up, we look forward to continuing the conversation in person at our Investor Day on May 21st. in Oregon. This will be an incredible opportunity to go deeper into the drivers behind our performance, hear directly from our customers in twist across faculty discovery and NGS workflow, and participate in a tool that brings our platform to life. You will also have the chance to engage with members of our management team as we discuss how we are scaling the platform, expanding into new applications, and driving long-term value creation. See you there. Thank you.

OPERATOR

Thank you for your participation. You may now disconnect. Good day.